

Market-Memory Mamba-RAG Trader

Retrieval-Augmented Temporal Modeling for Risk-Aware ETF Allocation

Data-driven Learning Term Project

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Abstract

The first Market-Memory RAG prototype was interpretable but too conservative: it matched raw short windows and often avoided market exposure. This project upgrades the idea into a neural retrieval system. A temporal encoder observes 120-day multi-asset ETF features, predicts five-day ETF returns, and produces a market-state embedding. Retrieval then checks whether similar historical embeddings led to consistent future outcomes. The final strategy uses a core-satellite portfolio rather than an all-or-nothing trading gate. In the historical test split, passive ETF baselines still win, but Mamba-RAG improves over Mamba-only and makes the signal auditable through historical analogues. This is a historical backtest, not investment advice.

1 Problem and Motivation

Daily return forecasts are weak. In investment settings, the more practical question is not simply “will the asset rise?” but “is the forecast trustworthy enough to allocate capital after costs and risk?” Raw nearest-neighbor retrieval is attractive because it is interpretable, but a 60-day price pattern is not a full market regime. We therefore combine a temporal neural encoder with retrieval-based market memory.

Market-Memory Mamba-RAG Pipeline

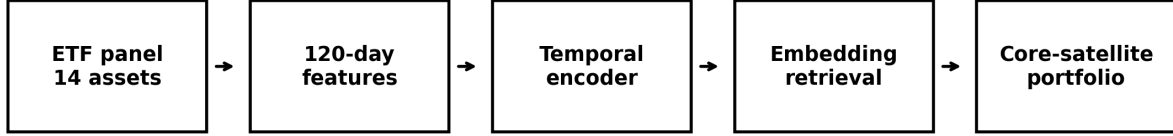


Figure 1: Pipeline. The encoder produces both forecasts and market-state embeddings. Retrieval audits the forecast, then a core-satellite rule converts the signal into ETF weights.

2 Method

Data. The ETF universe includes SPY, QQQ, IWM, TLT, IEF, SHY, GLD, and major sector ETFs. Features include multi-horizon returns, realized volatility, moving-average gaps, drawdowns, and cross-asset context. The prediction target is the five-trading-day forward ETF log return.

Temporal encoder. The intended backbone is Mamba-style sequence modeling. Mamba is motivated by selective state-space models for long sequence modeling with linear scaling in sequence length [1]. If the official Mamba kernel is unavailable, the implementation falls back to a TSMixer-style PyTorch encoder. TSMixer is a reasonable fallback because it mixes information along time and feature dimensions for multivariate time-series forecasting [2].

Retrieval. For each test date t , the encoder returns a forecast vector $\hat{y}_t$ and an embedding z_t :

$$(\hat{y}_t, z_t) = f_\theta(X_{t-119:t}).$$

Neighbors are retrieved from past train/validation embeddings only:

$$N_k(t) = \text{TopK}_{i \in \mathcal{D}_{past}} \text{sim}(z_t, z_i).$$

The retrieved future outcomes summarize whether similar historical market states supported the model forecast. This follows the broader motivation of retrieval-augmented time-series forecasting: retrieved history can provide an external inductive bias beyond model parameters alone [3].

Core-satellite allocation. The first RAG prototype could disappear into cash. The upgraded portfolio instead uses

$$w_t = (1 - \rho)w_{core} + \rho w_{satellite,t},$$

where the core maintains diversified exposure and the satellite tilts toward assets supported by both neural forecasts and retrieved outcomes. Component ablations use the same 5 bps L1 turnover transaction-cost assumption.

3 Main Results

The central result is modest and honest: passive exposure won the test split, but the proposed method improves over Mamba-only and gives a cleaner audit trail than a pure neural forecast.

Strategy	Final	CAGR	Sharpe	Max DD	Short interpretation
Buy-hold SPY	1.640	0.119	0.638	-0.243	Best final equity
Equal weight ETF	1.399	0.079	0.629	-0.177	Strong passive diversification
Raw KNN retrieval only	1.361	0.072	0.607	-0.145	Strong but turnover-intensive
Embedding retrieval only	1.248	–	0.345	-0.277	Learned memory only
Mamba-RAG core-satellite	1.173	0.037	0.374	-0.202	Proposed balanced method
Mamba only	1.131	0.028	0.216	-0.268	No retrieval ablation

Table 1: Main historical test metrics. The proposed method should not be presented as market-beating alpha. Its main evidence is improvement over Mamba-only and a more auditable decision process.

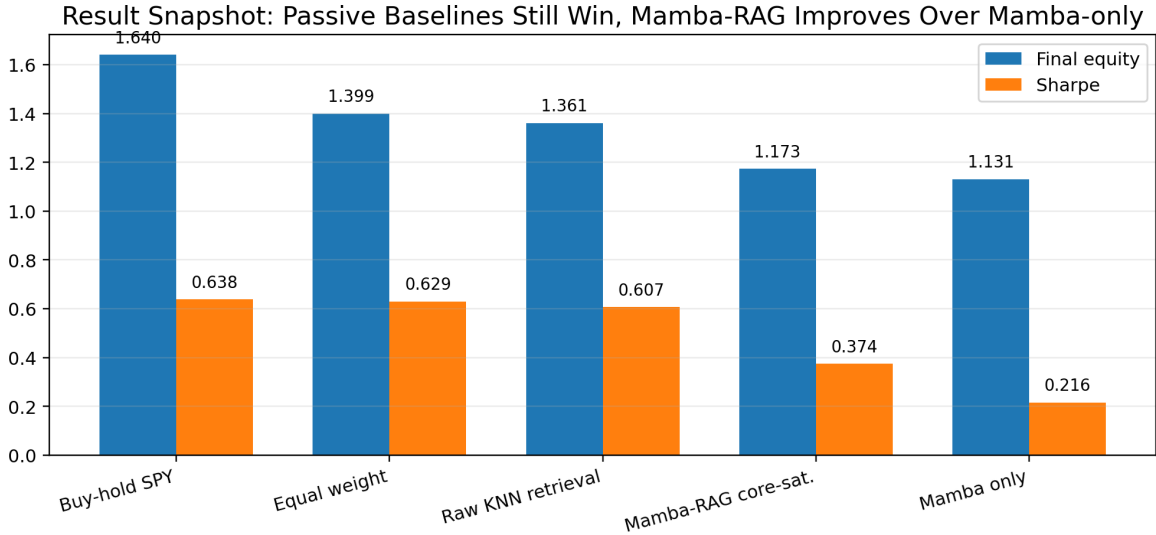


Figure 2: Result snapshot. Passive baselines remain difficult to beat; the proposed strategy’s clearest empirical win is over Mamba-only.

4 Component Ablation

The ablation directly addresses whether the RAG component matters. All four variants use the same test period and the same 5 bps L1 turnover cost.

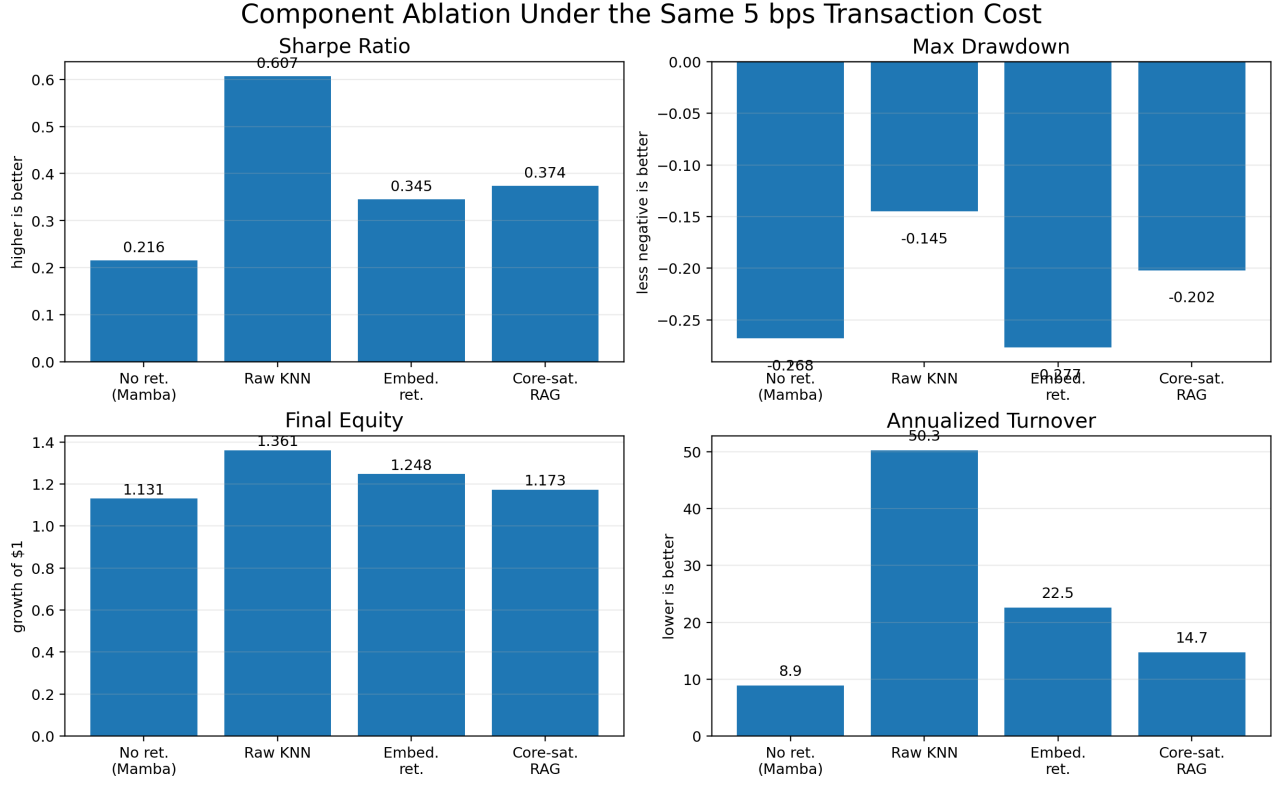


Figure 3: Component ablation. Retrieval improves over no retrieval. Raw KNN is the strongest standalone retrieval signal in this split, while the core-satellite design sacrifices some return for a more controlled allocation rule.

- **No retrieval:** Mamba-only isolates the temporal prediction signal and has the weakest Sharpe among the four component variants.
- **Raw KNN retrieval:** Strongest standalone retrieval result, but high annualized turnover makes it less attractive as a final portfolio rule.
- **Learned embedding retrieval:** Uses the learned market-state representation and improves over Mamba-only, making the neural signal more auditable.
- **Core-satellite Mamba-RAG:** A balanced allocation layer that avoids the old hard-abstention failure mode while remaining less aggressive than retrieval-only trading.

5 Diagnostics

The framework is useful because it is inspectable. We can see which assets were selected, when model and retrieval agreed, and whether embeddings form structured market-state trajectories.

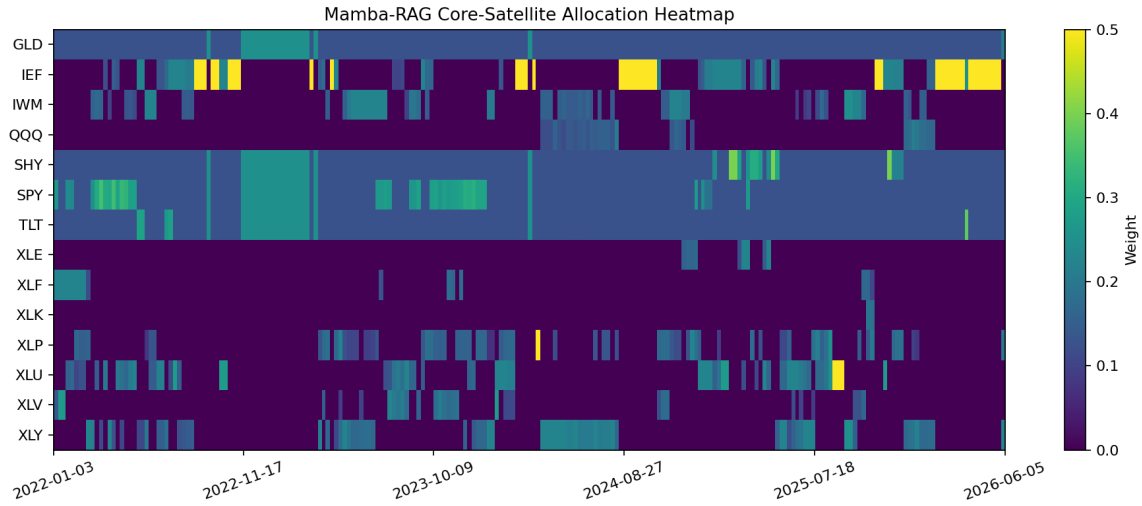


Figure 4: Allocation heatmap. Mamba-RAG maintains a core while tilting across bonds, gold, broad equities, and sectors.

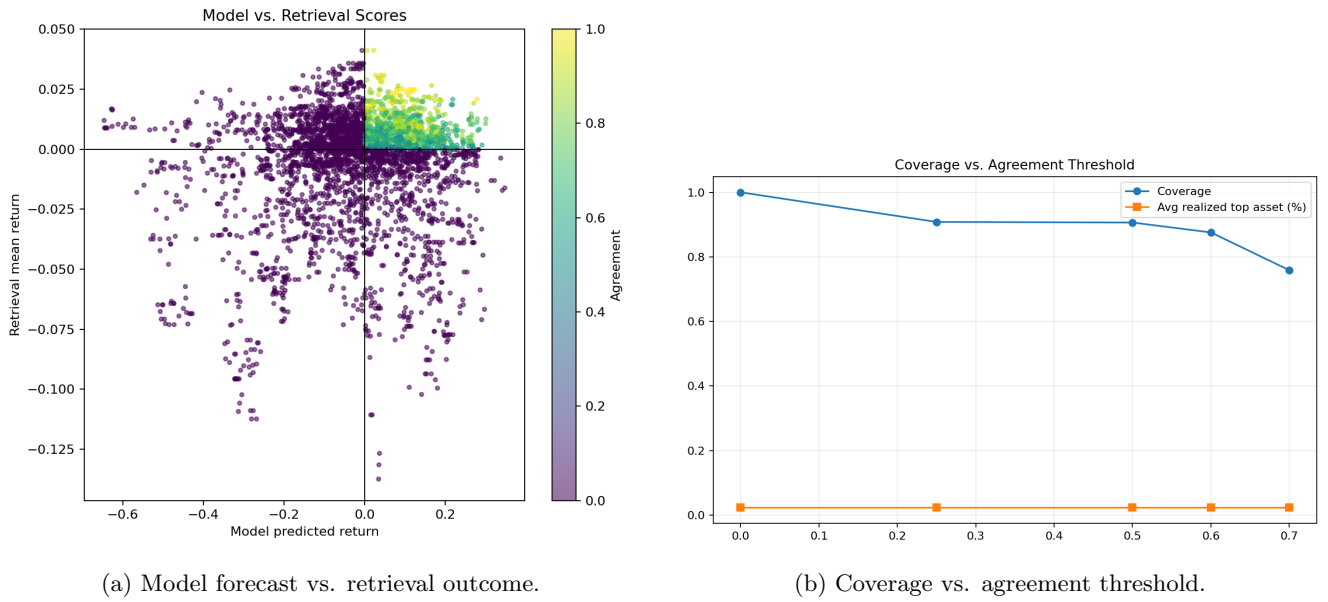
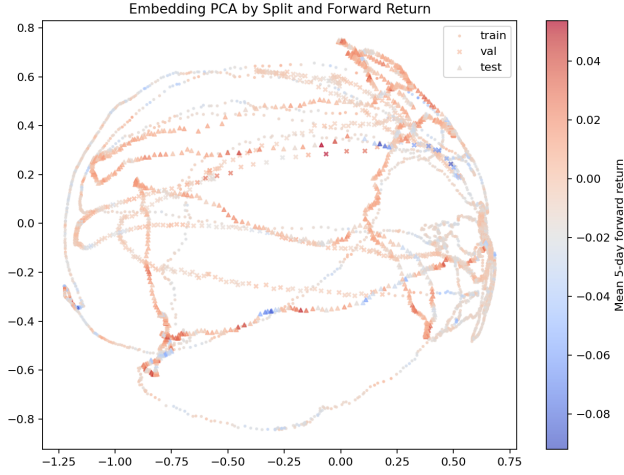
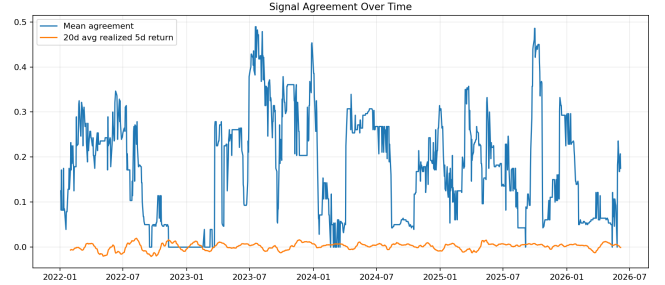


Figure 5: Retrieval diagnostics. The top-right quadrant in the score plot is where neural forecasts and retrieved historical outcomes agree. Raising the agreement threshold makes the strategy more selective.



(a) Embedding PCA.



(b) Agreement over time.

Figure 6: Additional sanity checks. The learned embedding forms structured trajectories, and model-retrieval agreement varies over time.

6 Limitations and Final Claim

This is a historical backtest with simplified costs, no taxes, no intraday execution, and no market-impact model. The evidence is not strong enough for a live trading claim. The next useful experiments are seed robustness, 60/120/252-day lookback ablation, and walk-forward retraining.

The appropriate final claim is:

Market-Memory Mamba-RAG demonstrates a structured way to combine temporal neural modeling with retrieval-based market memory. It improves interpretability and component behavior, but simple passive ETF baselines remain hard to beat.

References

- [1] A. Gu and T. Dao. *Mamba: Linear-Time Sequence Modeling with Selective State Spaces*. arXiv:2312.00752, 2023. <https://arxiv.org/abs/2312.00752>
- [2] S.-A. Chen, C.-L. Li, N. Yoder, S. O. Arik, and T. Pfister. *TSMixer: An All-MLP Architecture for Time Series Forecasting*. arXiv:2303.06053, 2023. <https://arxiv.org/abs/2303.06053>
- [3] S. Han, S. Lee, M. Cha, S. O. Arik, and J. Yoon. *Retrieval Augmented Time Series Forecasting*. arXiv:2505.04163, 2025. <https://arxiv.org/abs/2505.04163>